

Gravity at Altitude

Mayon Williams*

Georgia Institute of Technology, Atlanta, GA 30332

This paper develops a gravity-model Markov chain framework for ranking the structural importance of airports within an airline’s route network, using *Airline Enterprise* (AEn) – a persistent multiplayer airline management simulation – as a case study. The airline analysed, Pan-Himalayan Airlines (H3), operates 129 airports across five continents from its hub at Kathmandu (KTM). Passenger flow between any two airports is modelled as $\hat{D}(i, j) = C \cdot s_i^{\alpha_o} s_j^{\alpha_d}$, where s is airport size and C, α_o, α_d are fitted by maximum likelihood on a 60-route calibration sample obtained in-game. Two independent size measures are used: player-generated weekly passenger volume (in-game) and real-world annual passenger volume, yielding models with $R^2 = 0.745$ and $R^2 = 0.816$ respectively. Distance was found to be statistically insignificant in both fits and is instead treated as a sensitivity parameter $k \in [0, 1.5]$ governing a distance-decay term in the transition matrix; the resulting steady-state ranking transitions cleanly from a global-size- dominated ordering at $k = 0$ (Spearman $\rho = 1.00$ against raw size) to a geography-dominated ordering at $k \approx 1.0$ ($\rho \approx 0$). A secondary comparison of the two size measures across all 129 airports ($\rho = 0.939$) validates the game’s overall demand structure while identifying a set of major real-world hubs, including Bangkok (BKK), Guangzhou (CAN), and Jeddah (JED), that are substantially under-utilised within the current in-game economy relative to their real-world scale, representing actionable route-expansion opportunities for AEn players. A Zipf-law test on airport size versus in-game rank ($\beta = 1.199$, $R^2 = 0.692$) further indicates that AEn’s airport size distribution approximates, but is somewhat steeper than, the classic Zipfian power law. The resulting model and rankings are made available as an open-source, browser-based interactive tool for use by the wider AEn player community.

I. Nomenclature

N	=	number of airports in the network (129)
P	=	$N \times N$ Markov chain transition matrix
P_{ij}	=	transition probability from airport i to airport j
π	=	steady-state distribution vector (centrality scores)
$d(i, j)$	=	great-circle distance between airports i and j , nautical miles
s_i	=	size variable for airport i (weekly or annual passengers)
C	=	gravity model scale constant (fitted by MLE)
α_o	=	gravity model origin-size exponent (fitted by MLE)
α_d	=	gravity model destination-size exponent (fitted by MLE)
k	=	gravity model distance-decay exponent (sensitivity parameter)
$\hat{D}(i, j)$	=	gravity model predicted passenger flow from i to j
ε	=	log-normal multiplicative error term in the gravity model
σ	=	standard deviation of $\log(\varepsilon)$, estimated from residuals
R^2	=	coefficient of determination for gravity model fit
ρ	=	Spearman rank correlation coefficient

*Graduate Research Associate, Aerospace Systems Design Laboratory, Daniel Guggenheim School of Aerospace Engineering.

II. Introduction

A. Overview

The route networks of airlines are complex, varied, and fascinating systems to investigate for emergent behaviours. Among airports an airline formally serves, some nodes in the route network are structurally more central than others: they are visited more often by a "ghost passenger" – a hypothetical traveller making weighted-random connections; they attract more indirect traffic (passengers in transit); and they offer more leverage for route expansion.

This project aims to identify, given a set of airports, the ones that are most "central" by this definition, not just from a single airline's operational choices, but rather from the structure of the network itself.

This report presents a methodology for computing a centrality ranking for the set of airports in any airline's route network using a gravity-model Markov chain framework. The method requires only airport coordinates, a surrogate or proxy for airport size or demand, and a calibration sample set of origin-destination pair demand observations. The output is a ranked list of airports by steady-state network importance, derived from the underlying geography and market structure of the destination set, independent of the airline's own routing decisions.

B. Airline Enterprise

Airline Enterprise (AEn) is a persistent, browser-based real-time multiplayer airline management simulation in which players build and operate virtual carriers competing for passengers across a shared game world. Players lease gates, acquire aircraft, create routes, set fares, and compete for market share on shared routes. Each airport carries an in-game *rank*, a percentile statistic assigned by the developers and anchored to real-world passenger statistics*; this is the one size-related variable in the game that is solely tied to real-world data. Route-level demand, by contrast, is partly an emergent property of the simulated economy: while it depends to a certain extent on the 'size' of the origin and destination airports – as described above in relation to annual passenger volume – it also depends on how many players are actively flying that given route, at what frequency, and with what capacity, all interacting with the underlying rank-based ceiling. Market share at each airport is publicly visible, which enables cross-airline competitive analysis. Since *Airline Enterprise* is a shared world with economic consequences for routing decisions, a data-driven framework for identifying underserved high-centrality airports would likely be a useful decision-support tool for its player community.

C. Pan-Himalayan Airlines

The author operates an airline in the game called Pan-Himalayan Airlines, whose virtual IATA code is H3, and is henceforth referred to as both H3 and Pan-Himalayan. Based at its sole hub at Kathmandu's Tribhuvan International Airport (KTM/VNKT) in Nepal, the airline operates a network of routes across major airports around the world, as well as several smaller regional airports in the Himalayan region, South Asia, Southeast Asia, and Central Asia. Several airports served by H3 are substantially larger than KTM in both real-world and in-game statistical terms – for example London Heathrow (LHR), Atlanta (ATL), Dubai (DXB), and Singapore (SIN) – while several others in the network are considerably smaller. This heterogeneity, combined with H3's strict hub-and-spoke routing (all flights either originate or terminate at KTM, since AEn does not permit fifth-freedom flights), motivates the network-independent approach taken in this project: a centrality measure computed directly from H3's flown routes would trivially reveal the hub-and-spoke shape of the network rather than indicating anything about the underlying market. It may be noted that the majority of the routes flown by H3 are within Asia (as can be seen from Figure 1). This observation will, as it so happens, be important, as explained later in this report.

*The data used for this analysis is current as of July 2026



Fig. 1 Pan-Himalayan’s Route Map as of July 2026, with destinations outside Asia shown inset in the bottom-left corner.

D. Data Sources and Collection

The data sources used in this study are as follows: first, airport geographic coordinates (latitude and longitude) are drawn from the OpenFlights [1] open airport database and Google Maps (when data was not available in the OpenFlights database). These coordinates are used to compute great-circle distances between all airport pairs using the Haversine formula.

Second, two independent size measures are collected for each of the 129 airports in H3’s network:

- 1) ***In-game weekly passengers.*** H3’s own weekly passenger volume and its market share percentage are recorded at each airport from AEn’s in-game Airports Research page; the ratio of the two yields an estimate of total weekly passenger traffic at that airport. This quantity is an emergent **player-generated** statistic: it reflects the sum of passengers carried by every airline currently serving that airport, and not a statistic assigned directly by the developers.
- 2) ***Real-world annual passengers.*** Publicly available real-world annual passenger statistics for the corresponding physical airport, used as an independent, **non-player-generated** size measure.

The in-game airport rank (a percentile score assigned by the developers and tied to real-world data) is also recorded, both as a size proxy in its own right and as the basis of a Zipf-law validation check (Sec. V.C).

Third, a calibration sample of 60 origin-destination route pairs is assembled from the game’s Route Research tool, which reports the maximum capturable passenger load for any queried city pair at a standardised 100% route awareness, 100% airline reputation, and default ticket price. These 60 pairs are selected algorithmically to span the full range of distances (133–7,139 nmi) and airport size tiers present in the H3 network, while not necessarily being flown by H3 or any other airline in the game. H3’s hub at Kathmandu is deliberately under-represented (only 1 of 60 pairs touches KTM) to prevent the calibration sample from encoding H3’s own hub-and-spoke routing into the fitted model parameters.

III. Methodology

A. Gravity Model Specification

The maximum capturable passenger flow between any ordered airport pair (i, j) is modeled using a gravity specification analogous to Newton's law of gravitation: larger airports generate more traffic, whether as an origin or a destination. Formally,

$$\hat{D}(i, j) = C \cdot s_i^{\alpha_o} \cdot s_j^{\alpha_d} \cdot d(i, j)^{-k} \quad (1)$$

where C , α_o , α_d , and k are parameters that are estimated from the collected data [2]. Using log-transformed values of the data collected yields a linear model,

$$\log \hat{D}(i, j) = \log C + \alpha_o \log s_i + \alpha_d \log s_j - k \log d(i, j), \quad (2)$$

so that maximum likelihood estimation under a log-normal multiplicative error assumption reduces to ordinary least squares on the log-transformed variables.

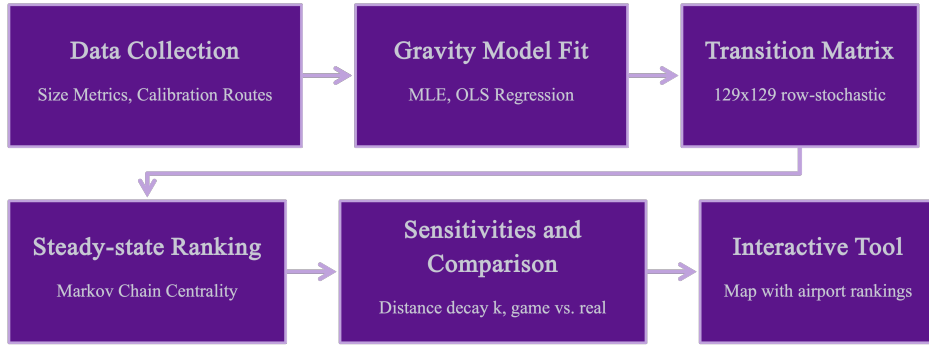


Fig. 2 Overview of the gravity-model Markov chain pipeline.

B. Maximum Likelihood Estimation

Parameters were estimated via OLS on the 60-pair calibration sample, separately for each size measure (in-game statistic vs. real-world). In both cases, the coefficient on $\log d(i, j)$ was statistically insignificant (Table 2), and the term was subsequently dropped from the fitted model. This is consistent with a demand structure in which, beyond a short-haul threshold, capturable demand depends only on the size of the two route endpoints rather than the distance between them. AEn's game guide confirms that flights below approximately 150 statute miles experience a demand penalty due to their competition with ground transportation, while longer flights are not described as subject to any further distance-based penalty [3]. The calibration sample, whose shortest pair is 133 nmi (153 mi), contains only a single observation near this threshold and is otherwise dominated by medium- and long-haul pairs where such an effect would not be expected to appear. Distance is therefore retained in the model as a theoretically-motivated sensitivity parameter k (Sec. V.D) rather than as an estimated coefficient.

Strictly speaking, the absence of a continuous distance-decay effect beyond the short-haul threshold means the fitted model is better characterised as a mass-interaction model with a discrete short-haul penalty than a classical continuous-decay gravity model. However the term 'gravity model' is retained here in its broader sense, consistent with common usage in transportation literature, but this distinction is worth flagging explicitly.

C. Transition Matrix Construction

For a chosen value of k , the fitted gravity model (with C , α_o , α_d at their estimated values from the collected data) is applied to all $N(N - 1)$ ordered pairs in the 129-airport network to produce a raw flow estimate $\hat{D}(i, j)$ for every pair – not just the 60 pairs used for calibration. Each row of the transition matrix P is then obtained by normalizing the raw outbound flows from airport i to sum to one:

$$P_{ij} = \hat{D}(i, j) / \sum_{l \neq i} \hat{D}(i, l). \quad (3)$$

Diagonal entries are set to zero (no self-transitions). The resulting matrix is stochastic by row and defines a finite discrete-time Markov chain over the 129 airports.

D. Steady-State Computation

The steady-state distribution π is the unique left eigenvector of P satisfying $\pi P = \pi$, $\sum_i \pi_i = 1$. Because P is constructed from the gravity model applied to *every* ordered pair, every airport has non-zero flow to and from every other airport by construction; the resulting chain is therefore irreducible and aperiodic [4], which guarantees the existence and uniqueness of π as well as convergence to it from any initial distribution. In practice, π is computed by power iteration, repeatedly applying $\pi \leftarrow \pi P$ until the L^1 norm of the update falls below 10^{-11} ; convergence is reached in under 40 iterations for all k values tested.

E. Validation

Model fit is assessed via in-sample R^2 , a 10-pair holdout validation, and normality tests on the calibration residuals. As a secondary check, the relationship between in-game airport rank and passenger size is tested against Zipf’s law (Sec. V.C), and the two independent size measures (in-game vs. real-world) are compared directly via Spearman rank correlation (Sec. V.E).

IV. Description of Data

The final dataset comprises 129 airports spanning five continents, with **in-game weekly passenger volumes** ranging from 340 (Neom Bay, NUM) to 2,448,000 (Singapore Changi, SIN), and **real-world annual passenger volumes** ranging from approximately 14,500 (Sittwe, AKY) to over 106 million (Atlanta, ATL). The 60-pair calibration sample spans distances from 133 nm (Jorhat – Shillong) to 7,139 nm (El Paso – Paro), with only one pair (Istanbul – Kathmandu) involving the H3 hub.

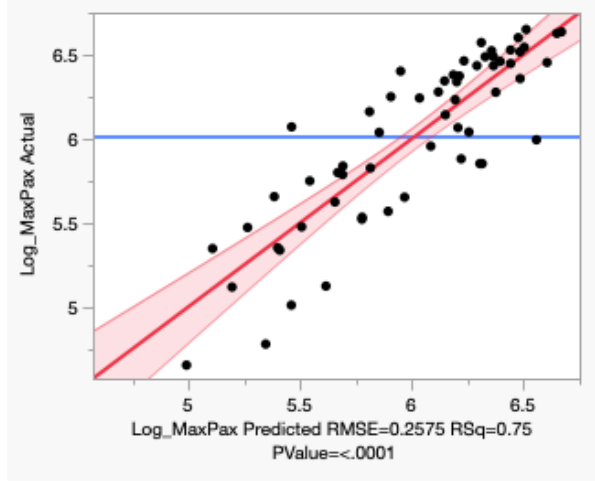
V. Results

A. Gravity Model Fit

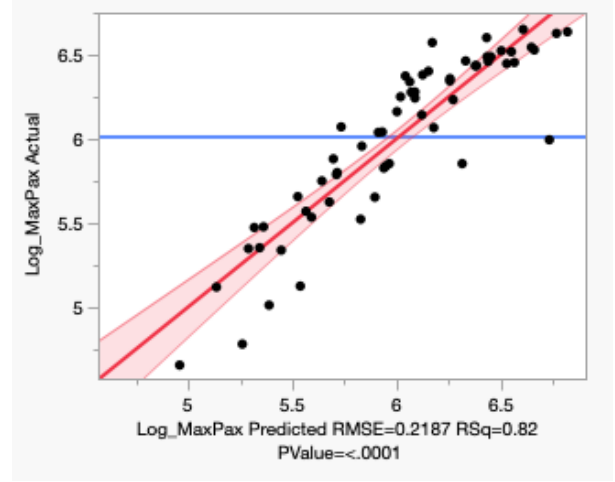
Table 2 reports the fitted parameters for four model specifications: with and without the distance term, for each of the two size measures. **Full** refers to models *with* the log-distance factor, and **partial** refers to model *without* it.

Table 2 Gravity model fit results (OLS on log-transformed calibration sample, $n = 60$)

	In-game (full)	In-game (partial)	Real-world (full)	Real-world (partial)
R^2	0.7509	0.7453	0.8173	0.8163
RMSE	0.2569	0.2575	0.2200	0.2187
C	30.60	37.75	9.47	10.07
k (p -value)	0.046 (0.267)	—	0.020 (0.568)	—
α_o	0.1118	0.1166	0.1215	0.1239
α_d	0.0948	0.1017	0.1210	0.1246



(a) In-game partial model.



(b) Real-world partial model.

Fig. 3 Actual vs. predicted $\log \hat{D}(i, j)$ for both size variant models.

Three findings stand out. First, distance is consistently insignificant across both size measures ($p = 0.267$ and $p = 0.568$), and dropping it costs negligible fit ($\Delta R^2 < 0.001$ in both cases), thus justifying the decision to treat k as a sensitivity parameter (Sec. V.D) rather than an estimated coefficient. Second, the real-world size measure fits the calibration data noticeably better than the in-game measure ($R^2 = 0.816$ vs. 0.745), which may indicate that AEn's route-level demand is more dependent on real-world passenger volumes at origin and destination airports than the current, player-driven state of the in-game economy. Third, the origin and destination exponents are nearly symmetric in both models, and *exactly* symmetric within rounding error for the real-world model ($\alpha_o = 0.1239$, $\alpha_d = 0.1246$), indicating that airport size contributes to passenger flow in an origin/destination-independent manner, i.e. larger airports generate more traffic whether they are the origin or destination, and to more or less the same degree in either position.

B. Model Validation

A 10-pair holdout validation (in-game partial model) produced residuals (mean -0.28 , range -0.66 to 0.40) of comparable magnitude and sign distribution to the in-sample residuals, indicating no evidence of overfitting given only three estimated parameters, though the small holdout size limits the precision of this check. A Shapiro–Wilk test on the full-sample ($n = 60$) residuals of the in-game partial model did not reject normality at the 5% level ($W = 0.9608$, $p = 0.0517$). Since ordinary least squares coefficient estimates do not require normally distributed residuals to remain unbiased, and because the intercept, α_o , and α_d all remain highly significant ($p < 0.0001$) in the in-game partial model regardless, this borderline result does not necessarily invalidate the model's core conclusions.

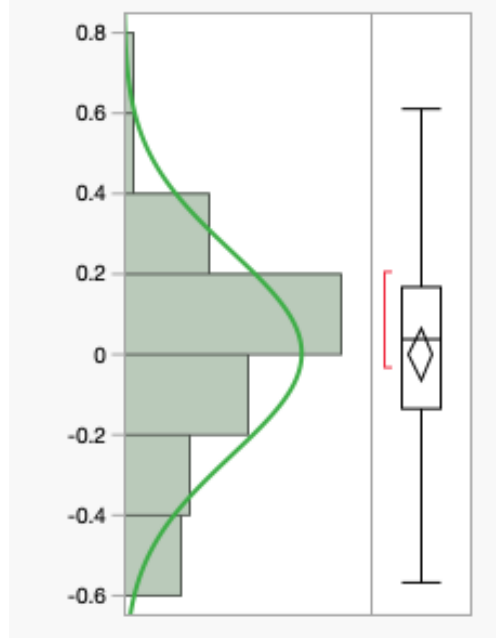


Fig. 4 Distribution of in-sample residuals with fitted normal curve.

C. Zipf-Law Check

As a secondary validation, the relationship between in-game airport rank and airport size (real-world annual passengers) was tested for consistency with Zipf's law [5], under which size scales as a power law of rank: $s \propto \text{rank}^{-\beta}$. A log-log regression yielded $\hat{\beta} = 1.199$ ($R^2 = 0.84$) (Figure 5), close to but somewhat steeper than the classic Zipf exponent of $\beta = 1$. This indicates that AEn's airport size distribution is approximately, though not exactly, Zipfian: the largest airports are proportionally larger relative to mid-tier airports than a pure Zipf law would predict.

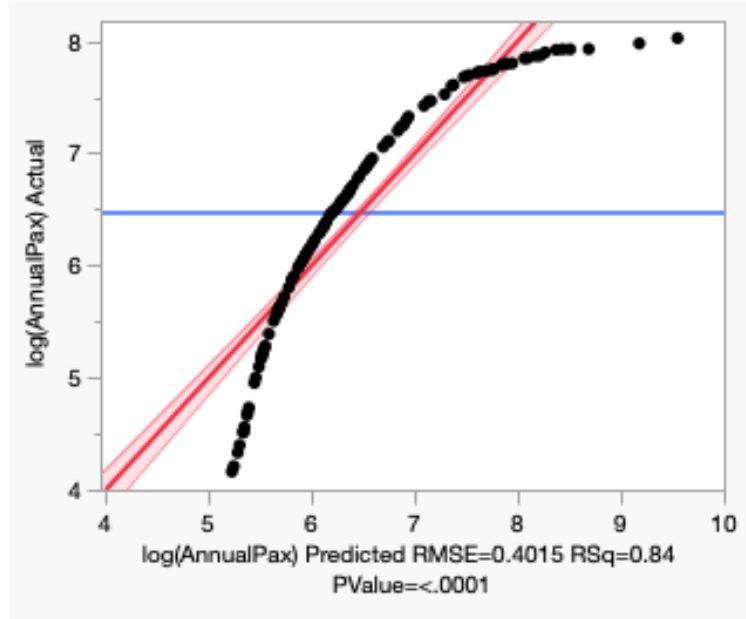


Fig. 5 Log-log relationship between in-game airport rank and real-world annual passenger volume.

In terms of gameplay, this means that the higher gate fees at a larger airport are, to a large extent, offset by the high

demand generated per-flight from these airports, thus making operating flights to/from larger airports a financially sounder decision than would necessarily be in the real-world.

D. Sensitivity Analysis on Distance Decay

Because the distance-decay exponent k could not be reliably estimated from the calibration data (Sec. III.B), the transition matrix and steady-state ranking were computed across a range of theoretical values, $k \in \{0, 0.5, 0.75, 1.0, 1.5\}$, using the in-game partial model. For each value, the Spearman rank correlation ρ between the resulting centrality ranking and the raw airport-size ranking was computed as shown in Figure 6, along with the resulting rank of the H3 hub, KTM, as well as a few other important airports in the region. (Table 3).

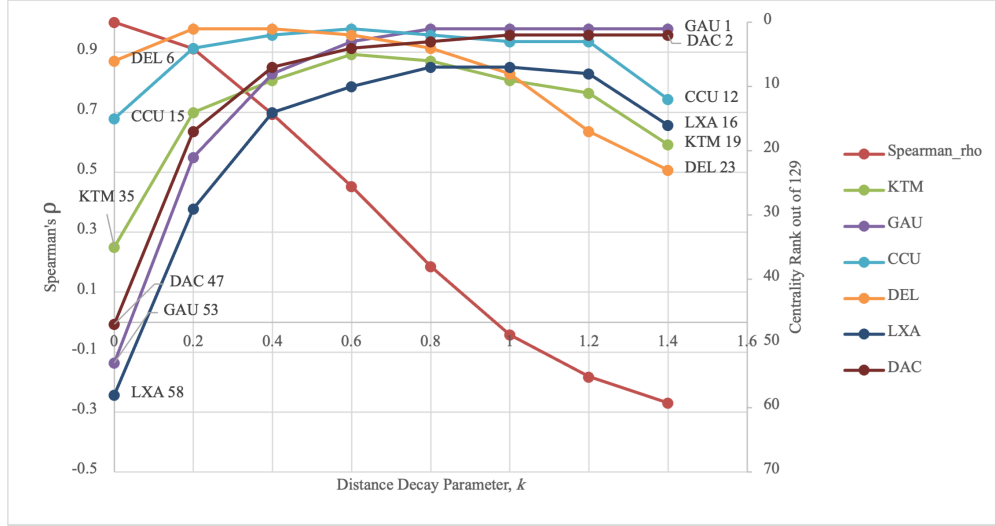


Fig. 6 Spearman correlation between centrality ranking and raw airport size, and six airports' centrality ranks, as functions of the distance-decay parameter k .

Table 3 Sensitivity of centrality ranking to distance-decay parameter k

k	0.0	0.5	0.75	1.0	1.5
Spearman ρ (vs. raw size)	1.000	0.579	0.244	-0.042	-0.305
KTM centrality rank (of 129)	35	8	6	9	21

At $k = 0$, the centrality ranking is by construction identical to a ranking by raw airport size ($\rho = 1$). As k increases, geography, particularly how close airports are to each other, increasingly reshapes the ranking; by $k \approx 1.0$ the correlation with raw size passes through zero, indicating a ranking that is essentially independent of airport size and instead reflects pure network position. Beyond $k = 1.0$, the correlation becomes negative, indicating that the model has begun to systematically favor small, geographically clustered airports over large, more isolated ones. Since this study only looks at a particular set of 129 airports out of the world's approximately 4000 functional airfields, at values of k beyond 1, the rankings become meaningless to airlines whose networks are not specifically focussed on the Himalayan region. The window $k \in [0.75, 1.0]$ is identified as the region in which geography and size both meaningfully contribute to the ranking without either dominating, and is adopted as the primary operating range for the interactive tool (Sec. V.H).

E. Comparison of In-Game and Real-World Size Measures

A direct comparison of the two independent size measures across all 129 airports yields a Spearman rank correlation of $\rho = 0.9389$, indicating that the aggregate, player-driven passenger volumes in AEn are strongly consistent with real-world airport traffic overall. This validates the use of either variable as a reasonable size proxy at the network level.

At the individual airport level, however, meaningful divergences are observed. Converting real-world annual figures to a weekly basis for direct comparison, Table 4 lists the airports with the largest positive divergence (in-game exceeds real-world) and the largest negative divergence (real-world exceeds in-game).

Table 4 Airports with largest in-game/real-world size divergence within study dataset

In-game > Real-world			Real-world > In-game		
IATA	Weekly (game)	Weekly (real)	IATA	Weekly (game)	Weekly (real)
MAA	2,048,000	407,831	DXB	685,000	1,830,769
HYD	1,768,000	560,893	BKK	307,500	1,209,657
SIN	2,448,000	1,345,769	ICN	637,333	1,424,451
BLR	1,768,000	783,269	JED	254,000	1,026,923
PEK	2,180,000	1,360,436	KMG	218,000	955,879
CMB	956,000	169,308	KUL	507,000	1,217,307
HKG	1,912,000	1,172,923	ATL	1,475,000	2,044,273
BOM	1,768,000	1,067,307	CGK	511,500	1,076,923
CCU	1,028,000	380,469	PVG	1,084,000	1,633,453
TRV	471,500	84,717	CAN	1,072,000	1,607,364

Since the in-game size measure is player-generated, this divergence is not a deliberate design choice by AEn’s developers. Instead, it reflects the current state of aggregate player activity relative to each airport’s real-world traffic level: airports in the left column are currently attracting disproportionately more flying activity from the player base than their real-world counterparts would suggest, while airports in the right column – several major Gulf and East/Southeast Asian hubs among them (DXB, BKK, ICN, JED, KMG, KUL) – are currently under-exploited relative to their real-world scale. The left column is dominated by South Asian and subcontinental airports, consistent with a player base concentrated in that region generating outsized reported passenger volumes there.

F. Steady-State Centrality Ranking

The steady-state ranking is presented at $k = 0.8$, the midpoint of the geography-dominant window identified in Sec. V.D, for both the in-game and real-world size measures (Table 5). The two rankings agree closely overall (Spearman $\rho = 0.917$ between the two full π vectors at $k = 0.8$; 12 of the top 20 airports are common to both), but diverge in interesting ways at the margin.

Both rankings are led overwhelmingly by South and Central Asian airports geographically close to KTM – consistent with the sensitivity analysis, which showed that at $k \approx 0.8$ the ranking is only weakly correlated with raw airport size and instead reflects network position within H3’s operating geography. The H3 hub itself, KTM, ranks 6th (in-game) and 8th (real-world): high, but not first, since centrality here measures structural importance to the whole 129-airport network rather than H3’s own operational reliance on the hub. The most notable divergence between the two rankings is the appearance of major Southeast and East Asian hubs (BKK, rank 9; CAN, rank 13; HKG, rank 19) in the real-world top 20 that do not appear in the in-game top 20 – these are airports whose true scale is only fully reflected once real-world passenger volumes are used as the size measure, a pattern examined further in the following section.

Table 5 Top 20 airports by steady-state centrality, $k = 0.8$

In-game model			Real-world model		
Rank	IATA	π (%)	Rank	IATA	π (%)
1	GAU	1.392	1	DAC	1.406
2	CCU	1.364	2	GAU	1.363
3	DAC	1.343	3	DEL	1.301
4	DEL	1.258	4	CCU	1.278
5	IXB	1.226	5	KMG	1.238
6	KTM	1.191	6	LXA	1.155
7	LXA	1.173	7	IXA	1.143
8	PBH	1.169	8	KTM	1.135
9	ISB	1.106	9	BKK	1.129
10	IMF	1.088	10	IMF	1.113
11	HYD	1.084	11	IXB	1.084
12	MAA	1.078	12	ZYL	1.082
13	SXR	1.070	13	CAN	1.078
14	BLR	1.067	14	CTU	1.072
15	CGP	1.067	15	CNX	1.069
16	LHE	1.035	16	HAN	1.054
17	TAS	1.027	17	CGP	1.053
18	CNX	1.026	18	IXS	1.028
19	BOM	1.007	19	HKG	1.025
20	KMG	0.998	20	ISB	1.023

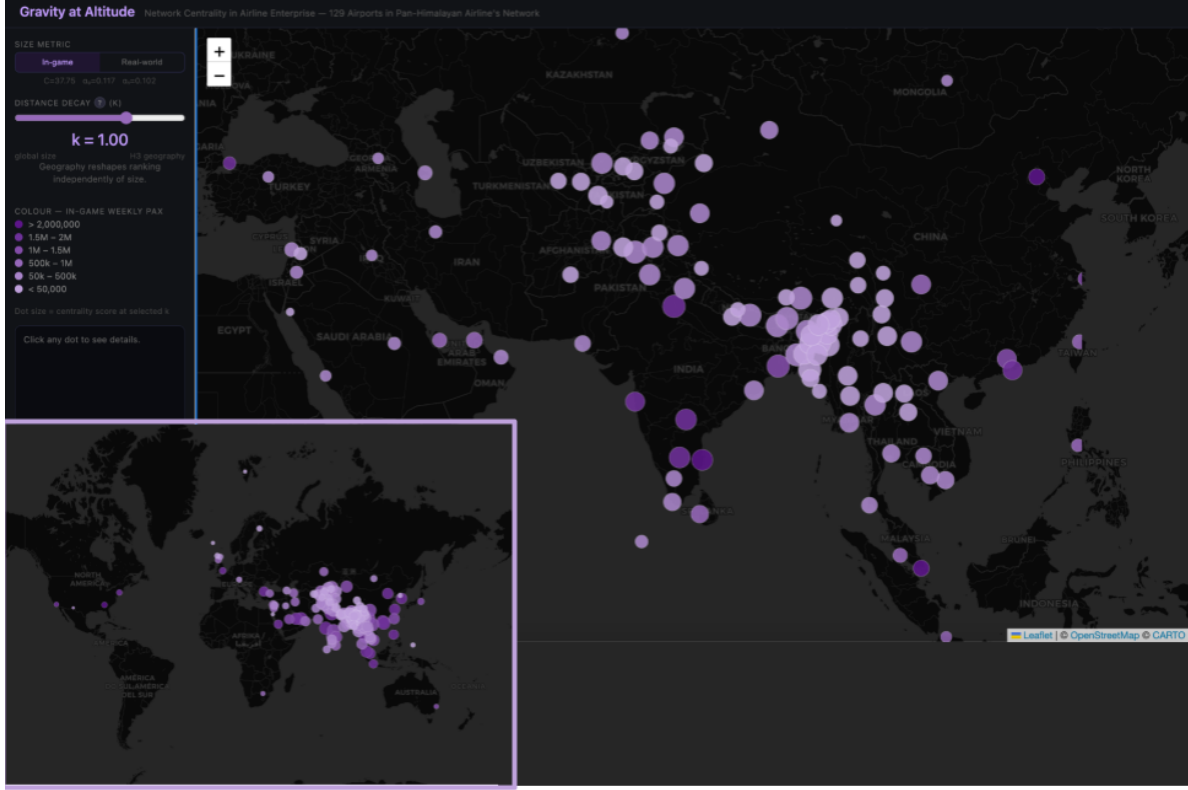


Fig. 7 Steady-state centrality across the 129-airport network, in-game model, $k = 0.8$. Marker size reflects centrality; colour reflects airport size tier. Airports outside Asia are shown inset in the bottom-left corner.

G. In-Game Under-Utilization of Large Real-World Hubs

The comparison in Sec. V.E identified a consistent set of airports for which real-world annual passenger volume substantially exceeds the current in-game weekly figure: Dubai Int'l (DXB), Bangkok Suvarnabhumi Int'l (BKK), Seoul Incheon Int'l (ICN), Jeddah King Abdulaziz Int'l (JED), Kunming Changshui Int'l (KMG), Kuala Lumpur Int'l (KUL), Guangzhou Baiyun Int'l (CAN), Jakarta Soekarno-Hatta Int'l (CGK), and Doha Hamad Int'l (DOH) all show real-world-equivalent weekly volumes 2–4× their current in-game figure. Cross-referencing these airports against H3's own market share reveals a consistent pattern: H3's utilisation at every one of these airports is below 1% (Table 6), despite several of them — BKK, CAN, and KMG in particular — ranking inside or near the top 20 of the real-world centrality ranking in Table 5.

Table 6 Major real-world hubs with disproportionately low H3 utilization

IATA	Real-world weekly-equiv.	In-game weekly	H3 market share
DXB	1,830,769	685,000	0.30%
BKK	1,209,657	307,500	0.80%
ICN	1,424,451	637,333	0.30%
JED	1,026,923	254,000	0.20%
KMG	955,879	218,000	0.50%
KUL	1,217,308	507,000	0.40%
CAN	1,607,364	1,072,000	0.10%
CGK	1,076,923	511,500	0.40%
DOH	1,044,231	511,500	0.40%

Because the in-game weekly figure is a player-generated statistic (Sec. 2.2) rather than a fixed ceiling, this pattern is best read as an indicator of currently under-exploited route potential rather than a structural limitation of the game world: real-world passenger volumes of this magnitude suggest that these airports could sustain substantially more flown capacity than the current aggregate player base is providing. For H3 specifically, and for any AEn airline able to serve these airports, this implies that flights to hubs such as BKK, CAN, and JED are likely to run at healthy load factors even at increased frequency or larger aircraft, since current in-game demand appears to be capacity-constrained by low player presence rather than by any inherent ceiling in the airport’s underlying size. This is probably the single most useful takeaway from this project for players of Airline Enterprise who want to capitalise on high-load factor, relatively low-competition destinations.

H. Interactive Tool

The centrality model, including both size measures and the full range of k values, has been implemented as a self-contained interactive web application. The tool displays all 129 airports on a world map, with marker size proportional to (rank-compressed) centrality and marker colour indicating airport size tier; a slider allows continuous adjustment of k from 0 to 1.5, with the underlying gravity model and steady-state distribution recomputed live in the browser at each value, and a toggle switches between the in-game and real-world size measures.

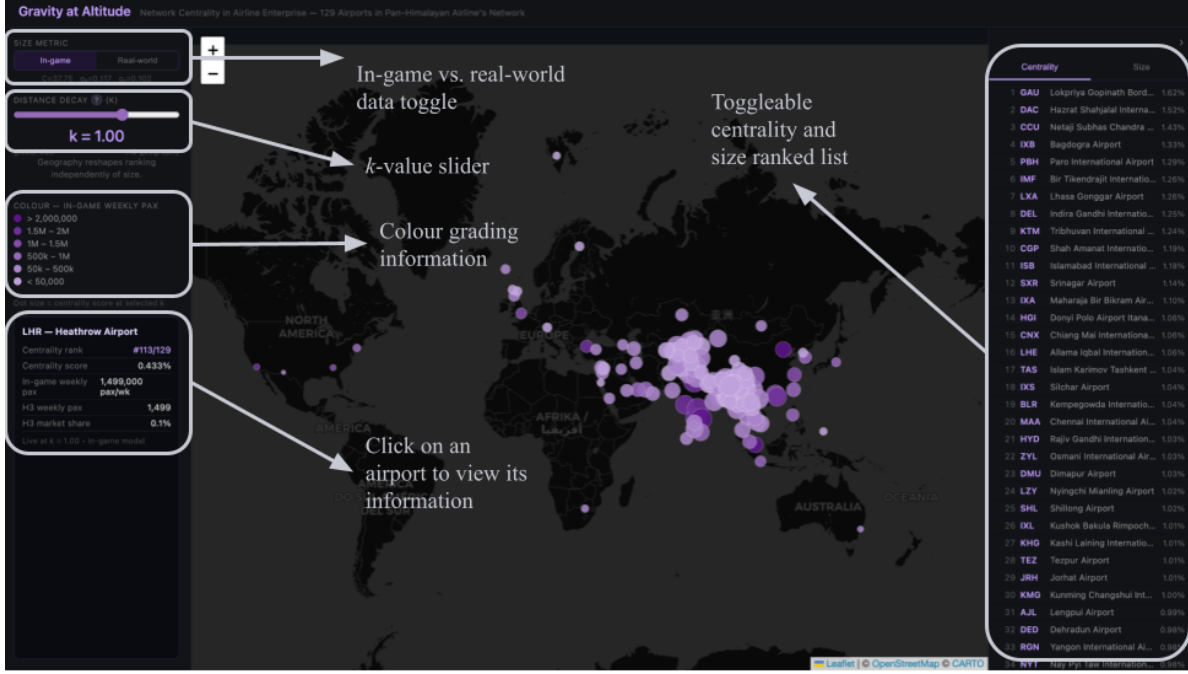


Fig. 8 The interactive web tool, showing the k -slider, size-metric toggle, and airport ranking panel.

The tool is intended for use by other AEn players evaluating route and hub-expansion opportunities. Should they wish, players are encouraged to copy the opensource code, Excel spreadsheet and add/substitute airports of their choosing to make the interactive tool more tailored to their own airlines. All the resources related to this study are available on the project’s GitHub page. The interactive tool can be accessed at https://mayonwilliams-source.github.io/Gravity_at_Altitude/.

VI. Conclusion

This report presented a gravity-model Markov chain framework for computing an airline-independent centrality ranking of airports within a simulated airline network, using Pan-Himalayan Airlines’ 129-airport route network in *Airline Enterprise* as a case study. Maximum likelihood estimation on a 60-route calibration sample yielded gravity models with $R^2 = 0.745$ (in-game player-generated size) and $R^2 = 0.816$ (real-world size), with distance found to be statistically insignificant in both cases, and is instead treated as a sensitivity parameter. The resulting steady-state ranking transitions from a global-size-dominated ordering at $k = 0$ to a geography-dominated ordering at $k \approx 1.0$ (Spearman ρ against raw size passing through zero), giving network operators a tunable lens through which to evaluate airports for either global reach or regional strategic centrality. A direct comparison of in-game and real-world airport size ($\rho = 0.939$) validates the game’s overall demand structure while surfacing a small set of airports where current in-game player activity diverges meaningfully from real-world traffic patterns.

The principal limitation of this study concerns the treatment of distance. The calibration sample provided no evidence of continuous distance decay beyond a short-haul threshold, consistent with AEn’s documented demand penalty for flights under 150 statute miles [3]; however, the sample contained only a single observation near this threshold (shortest pair: 133 nmi), and the game guide does not specify how, or whether, the penalty scales with distance below that cutoff. As discussed in Sec. III.B, this also raises an issue with terminology: since no continuous deterrence function is fitted, the model developed here is more precisely described as a mass-interaction model with a discrete

short-haul penalty than a classical gravity model in the strict Newtonian sense, and the term is retained chiefly for consistency with broader usage in the transportation literature. Future extensions of this work could recalibrate using a larger sample concentrated in the 50 to 150 mile range to characterise the shape of the short-haul penalty directly, extend the node set beyond a single airline’s destinations to test the sensitivity of the ranking to node-set selection, and incorporate a temporal dimension by tracking how centrality rankings and the in-game/real-world divergence evolve as the simulated economy develops.

Acknowledgments and Disclaimers

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Disclaimers

The in-game and real-world data used in this project is current as of late July/early August 2026. Additionally, as stated in the report above, the set of airports used in this study represent only a small fraction of total operational airfields in the world, and the conclusions from this report, while valid to a certain extent for large hubs both in real-world and in-game operations, does not apply to the vast majority of airports and airline networks. The work presented is simply a methodology by which others are encouraged to use to analyse other route networks and regions.

AI Use Declaration

The visualisation tool was developed with help from Claude (Anthropic).

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